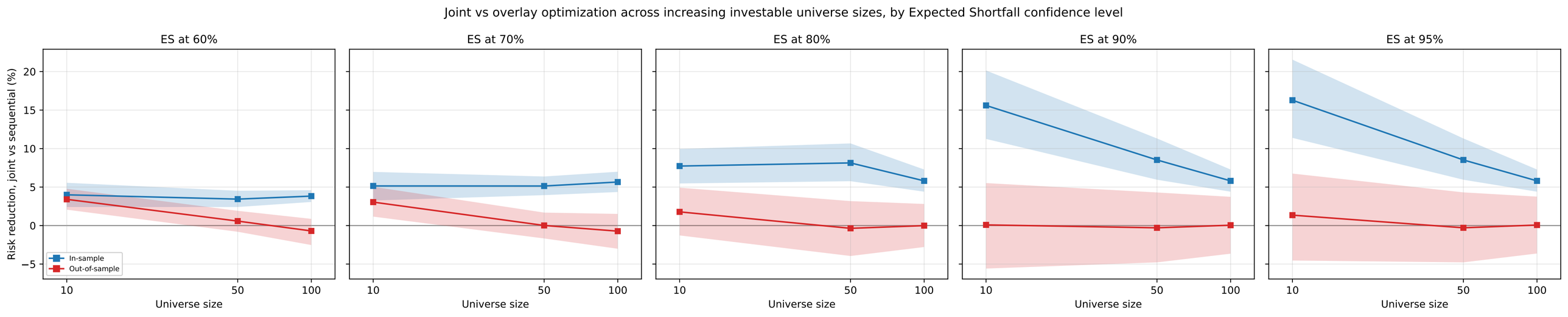
Joint vs overlay optimization across increasing investable universe sizes, by Expected Shortfall confidence level
ES at 60%
ES at 70%
ES at 80%
ES at 90%
ES at 95%
Risk reduction, joint vs sequential (%)
20
15
10
5
0
−5
In-sample
Out-of-sample
10
50
100
Universe size